

Day-Ahead Travel Time Forecasting on Freeway 5 Southbound with an Interpretable Linear Model

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Code and data: <https://github.com/leeshuncher/DataMiningHW1>

Abstract

Drivers from Taipei to Yilan decide the evening before when to leave, yet the Hsuehshan Tunnel bottleneck is congested only on certain days and hours. We forecast hourly southbound travel time on Freeway 5 (minutes, Nangang System to km 43.9) one day ahead from calendar and holiday-structure features alone, with a median linear regression of 109 terms whose coefficients read directly as minutes of delay. Trained on 2023–2024, with every choice made by rolling cross-validation, and tested on 2025, it reaches a test MAE of 1.42 min, 25% below last week’s same hour, and is statistically tied with gradient boosting (CV MAE 1.41 vs. 1.40, $p=0.72$). On long weekends, where the decision matters most, it halves the error (3.40 to 1.69 min, 95% CI 39–59%), and its recommended departure hour is within 0.82 min of the best hour, inside our 1.0 min decision threshold.

1 Introduction

Freeway 5 is the main fast route between Taipei and Yilan; its southbound direction funnels into the Hsuehshan Tunnel. Demand is mostly leisure travel, so queues follow the calendar: Saturday afternoons, the first day of a long weekend, the evening before a holiday.

User and decision. A driver (or trip app) chooses the evening before which departure hour to take tomorrow. Predictions are therefore made at 18:00 of the previous day from information known then.

Research question. Can a calendar-based linear regression forecast next-day southbound travel time well enough to choose a departure hour, and where does it fail?

Contributions. (1) A decision metric in minutes: the time lost by leaving at the recommended hour instead of the truly fastest hour in a 4-hour window, threshold 1.0 min. (2) A compact 109-term linear model whose coefficients are minutes of delay; it

Work	Data	Features/model	Horizon
Rice 04	Freeway travel times	current time, time of day; time-varying linear	short
Fan 18	Taiwan eTag	historical + real-time; ML	n/v
Bansal 20	Freeway 1, 36 km	43 predictors→6; BBO-SVR	n/v
Ashouri 23	Taiwan holidays 2019–21	Fourier terms; reconciled OLS	holiday patterns
Ours	Freeway 5 SB, 2023–25	calendar only; median linear, 109 terms	1 day

Table 1: Related work (n/v: not verified from sources we could open; errors not comparable across data).

ties gradient boosting in cross-validation, matches a ten-times larger one-hot model, and halves last week’s error on long weekends. (3) An analysis of what the model learned: long-weekend congestion is an afternoon effect (first-day mornings are *faster* than Saturday mornings), eve evenings are 12–29% slower, and the remaining large errors are incidents that no calendar can foresee.

2 Related Work

Table 1 compares four verified sources (checked on arXiv, the publisher or Crossref). Rice and van Zwet (2004) use a linear model with time-varying coefficients but need the current travel time; we have none the evening before. Fan et al. (2018) apply machine learning to Taiwan eTag data and Bansal (2020) applies support vector regression to a Freeway 1 segment; we study the holiday bottleneck of Freeway 5 with a readable linear model. Ashouri et al. (2023) analyse consecutive-holiday traffic with reconciled OLS forecasts and anomaly detection; we forecast travel time and evaluate the departure decision.

Split	Days	Rows	Mean	SD	>40 min
Train 2023	337	8,053	32.3	6.6	8.8%
Train 2024	366	8,679	31.5	4.8	5.0%
Test 2025	365	8,715	31.7	5.2	5.7%

Table 2: Target summary (minutes).

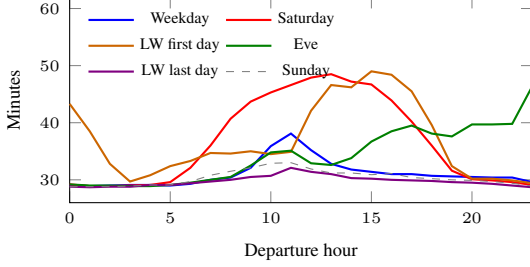


Figure 1: Mean hourly travel time by day type, 2023–2024 (LW: long weekend).

3 Task and Data

Target. Hourly mean small-car (type 31) travel time in minutes from Nangang System (gantry 05F0000S, km 0) to 05F0439S (km 43.9), southbound; free flow is about 30 min, the worst hour 87.8 min. One row is one departure hour; inputs are hour, weekday, holiday type and structure, known at 18:00 the day before. No same-day data or weather are used.

Data. TDCS M04A (median segment travel times per 5 min) for 2023-01-01 to 2025-12-31 (1,096 days, 26,304 hourly windows, 37 missing). We keep Freeway 5 southbound, type 31, valid rows only (seconds > 0, count ≥ 1), sum the four segment medians within the same 5-min slot (missing if any is invalid), and average to hours with at least 6 of 12 valid slots; nothing is imputed. The sum ignores that a 30-min trip crosses slots, and the endpoint is 2.6 km past the Toucheng exit. Holidays come from ruyut/TaiwanCalendar; a long weekend is ≥ 3 consecutive non-working days and the eve is the working day before. All methods are scored on identical rows (target and four weeks of history present).

4 Exploratory Analysis (training years only)

The shape of the day depends on the day type (Fig. 1): weekdays peak at 11:00 (38.1 min), Saturdays at 13:00 (48.5), the first day of a long weekend at 15:00 (49.0), and the eve rises late (46.4 at 23:00). The share of hours above 40 min is 3.7% on weekdays, 25.3% on Saturdays, 22.9% on first



Figure 2: Pipeline. Every learned quantity is fitted on 2023–2024 only.

Group	Terms	Hyp.
Hour of day	23 hourly main effects	–
Day type × block	9 day types (Sat, Sun, eve split Friday/other, long-weekend first/mid/last, single holiday, make-up day), each a main effect + 8 two-hour blocks 06–22h	H1,H2
Friday evening	4 terms (Fri × 14–22h blocks)	H3
Long holiday	1 term: first day, 12–18h, block ≥ 4 days	H4

Table 3: Feature groups; 23+81+4+1 = 109.

days. Saturdays drift: 30.3% in 2023 but 21.4% in 2024.

Association. Last week and the 4-week mean have the highest Pearson/Spearman/MI with the target (r 0.60–0.67, MI 0.74–0.80); hour of day has the highest MI among calendar variables (0.73). Holiday indicators look weak ($|r| \leq 0.14$) only because they cover few rows (10 first days and 10 eves in 703 days) yet their conditional effect is large; a correlation ranking would discard exactly the features that matter on long weekends (tested in §7). Redundancy: last week vs. 4-week mean $r=0.81$, Saturday vs. Saturday×daytime 0.62, so we check VIFs in the final model (max 3.1).

Hypotheses. H1: the first morning of a long weekend is much slower than an ordinary Saturday morning. H2: the last day barely affects southbound traffic. H3: the evening before a holiday is slower than an ordinary evening of the same weekday. H4: the first-day afternoon peak is lower for holidays of ≥ 4 days.

5 Method

Features. All features are 0/1 indicators of facts known the evening before, so every coefficient is minutes of delay (Table 3); after dropping empty columns the model has 109 terms. Last week, the 4-week mean and same-holiday-last-time have the strongest marginal association; adding them improved CV score by only 0.1%, below our 2% adoption rule, so they are excluded. Weather is unavailable.

Model. Median (L1) linear regression fitted by iteratively reweighted ridge (8 iterations). Two

properties of the data motivate it: the target is right-skewed (free flow 30 min, rare jams up to 88 min, Table 2), so a squared loss would let a few incident hours pull every coefficient, while the median is robust to them; and the indicators are only weakly collinear ($VIF \leq 3.1$), so strong shrinkage is unnecessary. Cross-validation confirmed L1 over a log-target least-squares fit (selection score 1.97 vs. 2.25). The ridge penalty α was selected from $\{0.01, 0.1, 1, 3, 10\}$ by 4-fold TimeSeriesSplit inside the training years; the tuning curve (CV MAE 1.661, **1.659**, 1.670, 1.741, 1.806, 1.903 for $\alpha=0.001, \dots, 10$; 2.213 at 100) shows that the chosen $\alpha=0.01$ is the minimum of an extended grid and that larger penalties over-shrink the rare holiday terms, so the model is essentially unpenalised. 2023 weekdays, Saturdays and Sundays get weight 0.1 because 2023 Saturdays were much more congested; holiday days keep weight 1. Selection score = mean of ordinary-day and holiday-day MAE; a compact design is eligible if within 5% of the best large one-hot model (2.001), and among eligible designs we take the lowest cross-validated departure loss. Blocks only: score 1.928, loss 0.715; blocks + hourly effects (chosen): 1.906, 0.304.

6 Experimental Setup

Protocol. Train 2023–2024, test 2025 (used for the final report). Rolling-origin CV on the four 2024 quarters, each trained on all earlier data, with the same folds, rows and weights for every method; we report mean $\pm$ SD. Significance: paired t -test over folds (low power) and a day-block bootstrap (2,000 resamples of whole days) on the test year; coefficient CIs use 300 day-resampled refits.

Metrics. MAE, RMSE, R^2 , and the decision regret: for each 4-hour window (06–10, 10–14, 14–18, 18–22h) recommend the hour with the lowest predicted time; regret = actual time there minus the window minimum. **Threshold:** mean regret ≤ 1.0 min on long-weekend days (a driver rarely notices less).

Baselines. Trivial training mean; last week same hour; same holiday last time (else last week); seasonal mean hour $\times$ weekday; simple OLS on last week; gradient boosting (L1 and log targets, LightGBM, same weights); a large one-hot ridge (over 1,000 hour $\times$ day-type terms).

Protocol note. The 2025 year was opened before the final design was fixed: a first compact design (two-hour blocks, chosen by MAE alone) showed a departure loss of 0.89 min, so we added the depart-

ure loss to the CV selection rule and hourly main effects. The re-selection itself used only cross-validation on 2023–2024; earlier log-target and weighting choices were likewise re-decided by CV (which reversed the loss choice), and all exploration tables use 2023–2024 only. We report this so that the 2025 numbers can be read accordingly.

7 Results

Main comparison. Table 4: on the test year our model is 25% (0.48 min, 95% CI 16–34%) below last week’s same hour, 50% below the training mean, and better in MAE than every simple baseline (trivial, persistence, seasonal mean, single-feature OLS), both in CV and on 2025. It has the best cross-validated holiday MAE of all methods (2.46 vs. 2.74 for GBM). Against the strongest baseline, GBM (L1), the CV difference is 0.01 min (paired t -test $p=0.72$, $n=4$); on the test year the two differ by 0.04 min (CI 0.01–0.08), 4% of the 1.0 min threshold, so in practice they are tied, and ours exposes every effect as a readable coefficient. The large one-hot ridge with over 1,000 terms is within 1.2% in MAE (CI -1.8 to 4.0%), so a design ten times smaller gives up no accuracy.

Where the gain comes from. The gain is largest exactly where the user’s decision is hardest: on long-weekend days MAE falls from 3.40 to 1.69 min (50%, CI 39–59%), and on ordinary days from 1.74 to 1.35 (22%, CI 12–32%).

Decision metric. Ours meets the threshold on long-weekend days ($0.82 < 1.0$ min; 88% of windows within 1 min of the best hour) and on all days (0.44; 92%). Choosing the hour by our forecast saves 1.09 min per window over leaving at the window start (CI 0.95–1.24), 1.54 on long weekends (CI 0.98–2.09) and 2.78 on first days (CI 1.44–3.92), and it more than halves last week’s long-weekend regret (1.81 to 0.82). GBM ranks hours better on first days (0.60 vs. 2.52, 8 days); so does the large one-hot ridge (0.65), which also has a separate term per first-day hour. Our design shares one value per two-hour first-day block, so it cannot order neighbouring hours inside a block: the gap is one of temporal resolution on 8 days, not of model family, and hourly first-day terms are the direct fix.

Feature ablation. Set A is hour, weekday and hour $\times$ weekday; B adds the 30 holiday/structure columns most correlated with the target (selected inside each training fold; 15 and 60 gave the same ranking); C is the full model. C is 15% better

Model	CV MAE	CV RMSE	CV hol. MAE	Test MAE	Test RMSE	Test R^2	Test hol. MAE
Trivial: training mean	3.00±0.48	4.80±0.84	3.76±0.18	2.82	5.21	0.00	3.52
Last week, same hour	1.73±0.37	4.60±0.84	3.01±0.47	1.90	4.99	0.08	3.23
Same holiday last time	1.75±0.32	4.66±0.78	3.21±1.48	1.80	4.77	0.16	2.34
Seasonal mean: hour×weekday	1.86±0.44	3.68±0.54	2.94±0.79	1.74	4.09	0.38	3.24
Simple: OLS on last week	2.22±0.45	4.08±0.64	3.24±0.08	2.10	4.39	0.29	3.19
Ours: compact median linear	1.41±0.23	3.79±0.66	2.46±0.19	1.42	3.93	0.43	2.00
Strong: GBM (L1)	1.40±0.25	3.77±0.72	2.74±0.03	1.38	3.93	0.43	1.83
Strong: GBM (log)	1.55±0.31	3.78±0.76	3.79±0.65	1.50	3.78	0.47	2.17
Flexible linear: large one-hot ridge	1.41±0.16	3.63±0.50	2.46±0.41	1.44	3.89	0.44	2.12

Table 4: Cross-validated (mean±SD over four folds) and 2025 test results, minutes; best in bold. Holiday = long-weekend days and eves.

Group	Start	Last wk	Ours	GBM
All days (1,455 win.)	1.52	0.66	0.44	0.36
Ordinary days	1.46	0.54	0.37	0.36
Long weekend days	2.36	1.81	0.82	0.33
LW first day (32)	5.30	4.50	2.52	0.60

Table 5: Mean regret (min per window), 2025. Start: leave at window start. Threshold 1.0 on long weekends.

than A (CI 9–21%) and 8% better than B (CI 6–10%) on the test year, with lower CV MAE as well (1.41 vs. 1.56 and 1.55). A correlation ranking is therefore not enough: it finds the holiday columns but not the day-type×block structure that tells the model *when* in the day the delay appears. The long-weekend terms are the most valuable group for the user: on first-day afternoons they cut error from 15.1 to 6.3 min (58%), and on long-weekend daytime hours from 5.5 to 3.0 min. Saturday and Friday terms carry ordinary days (9.0% and 2.7% of overall MAE). The ≥ 4 -day and eve terms change overall MAE by less than 1% (CIs include 0); we keep them because cross-validation selected them and because they encode H3 and H4, which §8 tests directly.

8 Analysis

Coefficients. Figure 3 shows every term with its interval and VIF bound, and Table 7 lists the key ones with p -values. Because all features are 0/1 indicators on the same scale, we report coefficients in minutes rather than per standard deviation: a standardised β would only rescale each term by its frequency and hide its meaning for the driver. The model learned a clear story. On weekdays the 10–11h peak adds 5.5–5.9 min, precisely estimated. Saturday 12–16h adds 12–15 min, and the first day of a long weekend adds 15–21 min at 12–18h, the largest effects in the model; their intervals are wide because the training years hold only about ten first

days, but all exclude zero. The last day of a long weekend is *faster* at 10–12h (−3.9 min), consistent with southbound demand having already left. The eve of a long weekend adds 8.3 min on Friday 20–22h and up to 14.6 min on other weekdays. Make-up working Saturdays add 11–21 min at 10–16h (6 training days): they behave like a busy Saturday, not like a working day. All VIFs are at most 3.1, so these estimates are not distorted by collinearity. **Hypothesis tests** (training years; OLS on log minutes, day-clustered SE). **H1 rejected, informatively:** first-day mornings are 18.4% *faster* than Saturday mornings, and the first-day excess appears only in the afternoon (+11.2%, $p=0.089$). Holiday traffic leaves later than weekend traffic, which is why the model needs day-type×block terms rather than a single holiday flag. **H2 mostly supported:** last-day effects are −0.3%, −0.6%, +0.4% in the morning, afternoon and evening, but 10–14h is 9.9% faster ($p = 0.0001$). **H3 supported:** eve evenings are +11.9% (Friday, $p < 0.001$) and +28.9% (other weekdays, $p = 0.0007$). **H4 suggestive:** −18.8% ($p = 0.028$ one-sided) in the log model, but the final-model interval covers zero.

Error analysis. The three largest 2025 errors are not holiday days: all are ordinary days under-predicted by about 44 min: 2025-07-01 10h (actual 79.1, predicted 34.7), 2025-08-16 (Sat) 11h (84.9 vs. 40.7) and 2025-09-12 11h (79.0 vs. 35.0); last week’s value was 35.0, 65.7, 32.4. In each, both the first segment (20–22 min) and the tunnel section (48–54 min) are slow. We have no incident, weather or road-work data, so the cause is unknown (incidents, closures or rain are plausible but unverified). The shared pattern is information the model never sees, not a calendar effect it mis-learned; the first fix is better inputs (weather forecasts, announced traffic-control notices), not a bigger model. Eve evenings, learned from about ten eves, are the remaining calendar slice to improve; a previous-year

Feature set	CV MAE	Test all	Ordinary	LW 08–18h	First day 10–18h	Eve 16–23h
A: calendar only	1.56±0.33	1.68	1.54	5.34	14.46	6.36
B: A + top-30 holiday columns	1.55±0.26	1.55	1.47	3.41	6.30	5.59
C: full	1.41±0.23	1.42	1.35	2.96	6.25	7.83
C – day-type×block	1.56±0.33	1.64	1.52	4.79	13.38	5.50
C – Friday terms	1.45±0.23	1.46	1.39	2.96	6.25	7.83
C – eve terms	1.41±0.23	1.41	1.35	2.96	6.25	6.59
C – long-weekend terms	1.44±0.27	1.52	1.35	5.49	15.14	7.83
C – long-holiday (≥4 d) term	1.41±0.23	1.42	1.35	3.00	6.44	7.83
C – Saturday terms	1.51±0.35	1.56	1.51	2.98	6.27	7.83

Table 6: Feature ablation, MAE in minutes (same L1 solver and weights). Slice columns are hourly MAE within the slice.

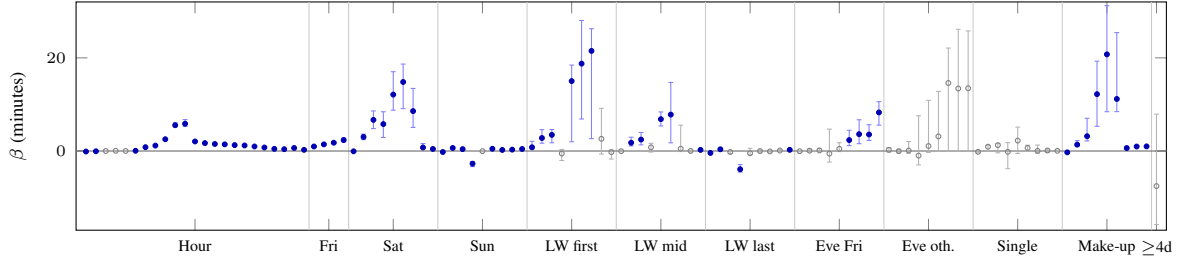


Figure 3: All 109 coefficients of the final model with 95% intervals (300 day-resampled refits). Filled: interval excludes 0 (69 of 109); hollow: includes 0. Within each day type the points run from the main effect through the eight two-hour blocks 06–22h. Every term is a 0/1 indicator, so β is in minutes and directly comparable across terms; all VIFs ≤ 3.1 .

Term	β	95% CI	p	VIF
Hour 10h	5.54	5.18, 6.05	.007	2.24
Hour 11h	5.86	5.30, 6.75	.007	2.24
Fri 20–22h	2.36	1.97, 2.78	.007	1.22
Sat 14–16h	14.82	9.09, 18.66	.007	1.51
LW first 14–16h	18.76	6.87, 28.02	.007	1.94
LW first 16–18h	21.49	2.66, 26.26	.013	1.94
LW first 10–12h	−0.54	−2.04, 0.33	.160	1.27
LW last 10–12h	−3.91	−4.52, −2.92	.007	1.28
Eve (Fri) 20–22h	8.28	5.55, 10.60	.007	1.27
Eve (other) 16–18h	14.60	0.00, 22.11	.007	1.26
First day, ≥4 d	−7.53	−15.79, 7.91	.293	2.99

Table 7: Selected coefficients (minutes; 300 day-resampled refits, so $p \geq 2/300 \approx .007$). Full table: `final2_coefficients.csv`.

same-eve feature is the next experiment.

9 Conclusion and Limitations

Answer to the research question: yes. Calendar and holiday structure alone let a 109-term median linear model forecast day-ahead hourly travel time with test MAE 1.42 min, 25% below last week’s same hour and 50% below it on long weekends, and its recommended departure hour is within 0.82 min of the best hour on long weekends, inside the 1.0 min threshold. It is tied with gradient boosting and a ten-times larger linear model in accuracy while every term remains a readable number of minutes.

What mattered was not *whether* a day is a holiday but *when* in the day each holiday type is slow: first-day and Saturday afternoons, eve evenings. *Recommendation:* a trip app can use the forecast directly to suggest departure hours before long weekends; for the first day, shifting a departure out of the 12–18h block saves the most time. *Limitations:* one test year and few holiday days (31 long-weekend days, 8 first days, 8 eves) give wide intervals; 2025 was not fully blind (see Protocol note); the target ends 2.6 km past the Toucheng exit and sums segment times; 2023 Saturdays differ and were down-weighted; no weather or incident data; the 1.0 min threshold is our assumption. Next: weather and traffic-control inputs, a same-eve feature, and 2026 data.

Contributions of Each Member

[NAME (STUDENT ID)]: [task]; [NAME (STUDENT ID)]: [task]; [NAME (STUDENT ID)]: [task].

AI Usage Statement

We used Claude (Anthropic) as a coding and writing assistant: it wrote the data-processing, modelling and evaluation scripts, ran the experiments and drafted this paper. The team chose the topic,

reviewed the protocol and results (a review found test-set leakage in an earlier version, which was corrected), and is responsible for the content. [Team: describe what each member checked.]

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